

**66 Lower Court Road
Epsom
KT19 8SN**

**Assessment of Known Defects
and
Recommendations on Remedial Works**

**Prepared for:
Stephen Hamilton
November 2024**

Prepared by Sam Heath
MSc MFPWS

Reviewed by Michael White
MRICS FCABE FFPWS

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TABLE OF CONTENTS

INSTRUCTIONS	3
DECLARATION	3
SCOPE OF WORKS	4
EXISTING STRUCTURES	5
SOIL CONDITIONS	6
DEFECTS IDENTIFIED	8
POTENTIAL CAUSES	13
RECOMMENDED FURTHER INVESTIGATIVE & REMEDIAL WORK	14
CONCLUSIONS	15
APPENDIX	16



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INSTRUCTIONS

White and Lloyd have been instructed by Stephen Hamilton to carry out a site inspection and produce a Structural Assessment on identified defects at the property known as **66 Lower Court Road, Epsom KT19 8SN**.

We were instructed via email on 29th October 2024. We carried out our site visit and inspection on 5th November 2024.

DECLARATION

Our team of engineering specialists offer advanced levels of expertise to our clients. We have a wide experience of the construction and development industry as well as producing structural assessments for residential dwellings and commercial properties.

This report has been made by Sam Heath MSc MFPWS. Sam is the senior building surveyor at White & Lloyd with over 7 years' experience in the construction industry.

This report has been checked and reviewed by Michael White FRICS C.Build.E. FCABE FFPWS. Michael is a chartered building surveyor, a chartered building engineer and fellow of the Chartered Association of Building Engineers. He has over 20 years' experience in the construction industry.

We certify this report is an accurate reflection given the information available. All statements contained within this report are based on professional opinion.



Sam Heath
Author



Michael White
Reviewer



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SCOPE OF WORKS

This report includes the following details:

1. A description of the existing building, site conditions and assumed substructure.
2. A description of the existing structural defects identified at the time of our site visit.
3. Potential causes of the defects identified.
4. Recommended remedial work.

This report is prepared following a visual inspection of the relevant areas of the site, as far as access was obtained. The report does not include detailed in-depth investigations to any parts of the premises, which are covered, inaccessible, or unexposed. No excavations were carried out as part of the investigations, and we have not inspected the substructure.

Important note: We carry out a visual inspection only. We inspect external walls and surrounding structures, from ground level and, if possible, from neighbouring public property. We do not lift the covers to inspection chambers, and we do not inspect the underground drainage system.



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EXISTING STRUCTURES

66 Lower Court Road is a two-storey, semi-detached house, featuring off-street parking at the front and a garden to the rear. The exterior walls to the main property are believed to be cavity, masonry walls with solid masonry walls to the single storey rear extension. The windows are uPVC double glazed units.

The roof is a cross pitched timber frame with natural slate coverings and angled ridge tiles.



Image 1: General view of 66 Lower Court Road, front elevation.



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SOIL CONDITIONS

Research through the British Geological Survey shows the following soil conditions of the site:

The bedrock geology consists of London Clay Formation - Clay and silt. Sedimentary bedrock formed between 56 and 47.8 million years ago during the Palaeogene period. See image 2 below.

The superficial geology consists of Kempton Park Gravel Member - Sand and gravel. Sedimentary superficial deposit formed between 116 and 11.8 thousand years ago during the Quaternary period. See image 3 below.



Geology



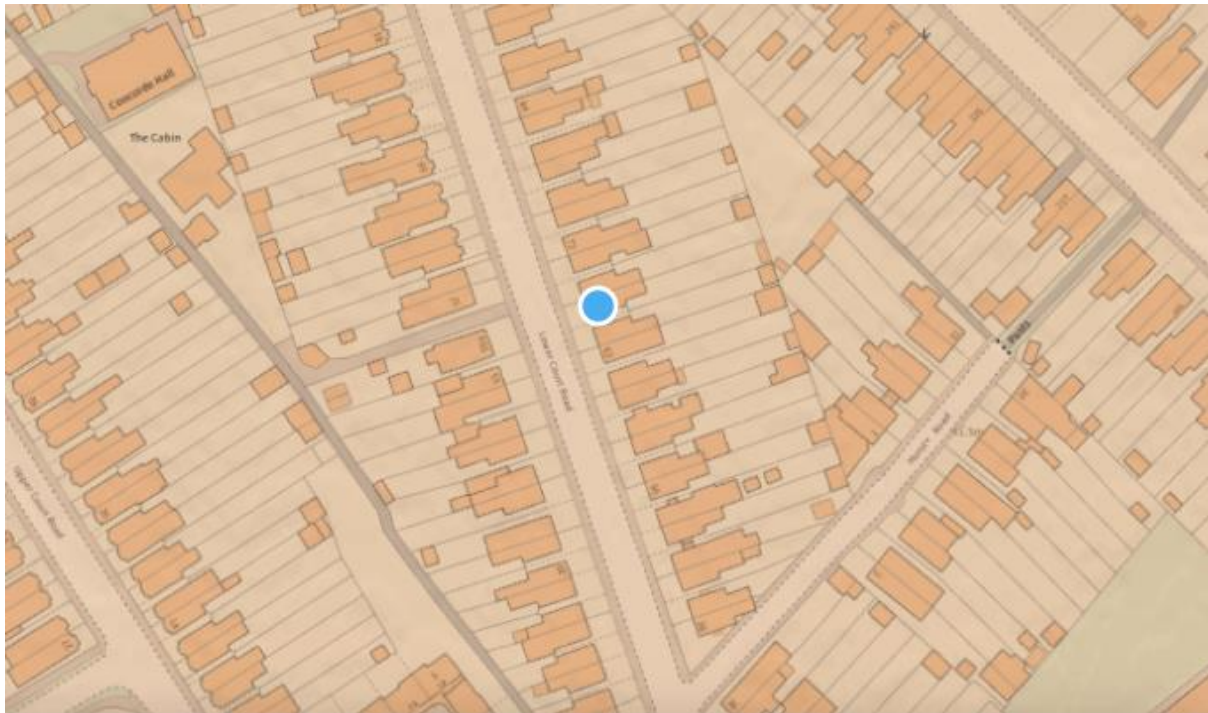
Bedrock geology

London Clay Formation - Clay and silt. Sedimentary bedrock formed between 56 and 47.8 million years ago during the Palaeogene period.

Image 2: Geological Map showing bedrock geology.
Contains British Geological Survey materials © UKRI-2024



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Geology



Superficial deposits

River Terrace Deposits - Sand and gravel. Sedimentary superficial deposit formed between 2.588 million years ago and the present during the Quaternary period.

Image 3: Geological Map showing superficial geology.
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DEFECTS IDENTIFIED

We inspected the internal and external rear elevation of the property. Our inspection identified the following areas of structural concern:

1. The rear flank chimney stack is leaning substantially inwards.
2. Evidence of defective and bulging brickwork to the end gable wall.

This section provides a description of each defect. A structural assessment and comment is provided in the following section. Photographic evidence of the defects noted above is provided on the following pages. Electronic photographs taken will be held on the surveyors file for a period of six years and are available on request.

Rear Flank Chimney Stack Leaning Substantially Inwards

The rear flank chimney breast, which protrudes above the roof structure for 21 courses of brickwork, giving it an approximate height of 1.8m above the roof structure, was observed to be leaning inwards significantly (Image 4). Additionally, there was evidence of moisture ingress at ground floor level within the masonry of the flank wall, adjacent to the rear flank chimney breast (Image 5).

Defective and Bulging Brickwork to the End Gable Wall

The external masonry to the end gable (flank) wall was noted to be bulging, notably towards the ridge line of the roof structure (Images 6 & 7). Within the internal loft space, missing, cracked and bulging brickwork was observed. There was also evidence of gaps in the roof finishes and moisture ingress within the internal loft space.



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Image 4: Rear flank chimney breast leaning inwards.



Image 5: Evidence of moisture ingress within flank wall masonry adjacent to rear flank chimney breast.



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Image 6: End gable (flank) wall out of level at top of wall, near to ridge line.



Image 7: End gable (flank) wall out of level at top of wall, near to ridge line.



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Image 8: Rear flank chimney breast within internal loft space.



Image 9: Bulged and missing brickwork within internal loft space. Wall tie circled.



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Image 10: Missing brickwork beneath front roof purlin.



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POTENTIAL CAUSES

The following section outlines the potential causes of the observed defects.

Rear Flank Chimney Stack Leaning Substantially Inwards

It is believed that a lack of support for the structure has resulted in the observed leaning of the chimney breast. From ground level, the external brickwork and mortar of the chimney breast appear in good condition, with no obvious defects. The chimney protrudes approximately 1.8m above the roof structure and there is no additional support above the roof structure, the lack of support has allowed the chimney breast to lean substantially inwards. A visual inspection of other properties on Lower Court Road indicated that this had occurred to the other properties too, with many having removed the rear flank chimney breast to below the roof structure. It is therefore assumed that the observed leaning of the chimney breast has been caused by a construction defect, namely the lack of any support above the roof structure.

Defective and Bulging Brickwork to the End Gable Wall

The bulging of the brickwork to the end gable wall is believed to have been caused by insufficient tying together of the inner masonry leaf and outer masonry leaf with wall ties. During the site visit, the presence of wall ties was observed, however it is believed that these wall ties are either defective through corrosion, or alternatively that there are not a sufficient number of wall ties to correctly tie the two structures together and prevent the observed movement. It was also observed that sections of masonry within the inner leaf were missing, meaning that there are no wall ties tying the structures together at these locations.



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RECOMMENDED FURTHER INVESTIGATIVE AND REMEDIAL WORKS

The following remedial works are recommended to address the defects observed at the property.

Rear Flank Chimney Stack Leaning Substantially Inwards

The chimney stack should be reduced in height to below the roof level as the chimney is no longer in use, with the internal fireplaces now covered. A roofing contractor should be instructed to remove the chimney stack to below the roof level, and to install new roof finishes to cover where the chimney stack currently protrudes from the roof structure, in order to prevent moisture ingress within the chimney breast. The reduced height chimney stack should be covered to prevent any materials or moisture getting into the disused flues. The reduction and covering of the chimney stack will also help to address the moisture ingress within the masonry of the flank wall that was observed during the site visit and will help to protect the external masonry, as this is believed to have been caused by the lack of capping on the chimney flues allowing water into the flues through rainfall.

It is expected that a planning application would be required to undertake the necessary remedial works to the chimney stack. A visual inspection of other properties on Lower Court Road indicated that the lowering of the chimney stack has been undertaken at other properties

Defective and Bulging Brickwork to the End Gable Wall

Any areas of defective masonry are required to be replaced with new masonry and any areas where missing masonry was observed to be missing, notably within the inner leaf within the internal loft space, should be filled with new masonry. A greater number of wall ties are required to correctly tie the inner leaf and outer leaf together and to prevent the bulging brickwork that was observed. Where new masonry is being installed, wall ties can be installed simultaneously. Where existing masonry is not being replaced, wall ties can be installed within the existing masonry. It must be ensured that any newly installed wall ties are correctly installed, and will therefore correctly tie the inner leaf and outer leaf together, preventing the observed bulging from occurring. Ideally, wall ties should be evenly distributed so that there are a minimum of 2.5 wall ties per square metre and they should be staggered between courses of brickwork.



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CONCLUSIONS

Our site visit on Tuesday 5th November 2024 was a non-intrusive investigation. The following conclusions are therefore made following a visual inspection.

- The rear flank chimney stack is leaning substantially inwards.
- The end gable (flank) wall is bulging at the top, close to the ridgeline of the roof structure.
- A construction defect, namely a lack of support of the 1.8m of chimney stack above the roof structure is believed to be the cause of the observed leaning.
- Failed wall ties, or an insufficient number of wall ties is believed to be the cause of the observed bulging brickwork to the end gable wall at high level.
- Within the internal loft space, sections of the inner leaf wall were observed to be missing. In these locations, there were no wall ties present.
- It is recommended to reduce the height of the chimney stack to a new level within the internal loft space.
- New roof finishes should be installed to cover where the chimney stack currently protrudes from the roof structure.
- The reduced height chimney stack should be covered to prevent ingress of materials or moisture.
- The reduction in height of the chimney stack is likely to require a planning application.
- A rough budget estimate has been provided (see Appendix A) for the remedial works.



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APPENDIX

The costs for the defects identified within this report are contained here. These costs are prepared as a budgetary exercise to provide information on the current costs of the defective items identified. The costs outlined are based on current construction rates as at Q4 2024. An allowance is then made for contractors overheads and other usual costs applied.

These costs do not allow for payments to design consultants (architectural or structural services). We have also not allowed for any payments to local authorities for statutory consents (planning / building control).

Item Description	Budget Costing
Scaffolding access to provide safe access to chimney stack above roof level and gable end repairs	£2,500
Waste removal, including skip hire and waste chute	£350
Roof infill, to include supply and fixing of timber rafters, roofing membrane, insulation, battens and tiled finishes, all to match existing.	£650
Masonry repairs to gable end, includes allowance to replace up to 4 sq.m of masonry. Includes installation of wall ties as required	£800
Preliminaries (site set up, H&S signage, site management), at 12%	£516
Contractors overheads and profit, at 15%	£645
VAT at 20%	£1,092
Total For Building Costs:	£6,553

Appendix A – Budget Estimate

****END REPORT****



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